Espacios Euclídeos.

① ¿ Prod. escalar ? $\Leftrightarrow$ ¿ G simétrica + def + ?

a) $x \cdot y = 3x_1y_1 + x_1y_2 + x_2y_1 + 2x_2y_2$

$$= (x_1, x_2) \begin{pmatrix} 3 & 1 \\ 1 & 2 \end{pmatrix} \begin{pmatrix} y_1 \\ y_2 \end{pmatrix}.$$

Como $G = \begin{pmatrix} 3 & 1 \\ 1 & 2 \end{pmatrix}$ es simétrica y def. pos:

$$|3| = 3 > 0$$

$$\begin{vmatrix} 3 & 1 \\ 1 & 2 \end{vmatrix} = 5 > 0$$

$\Rightarrow$ $x \cdot y$ define un prod. escalar.

b) $x \cdot y = x_1y_1 + x_1y_2 + x_2y_1 + x_2y_2$

$$= (x_1, x_2) \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} y_1 \\ y_2 \end{pmatrix}$$

Simétrica ✓

Def. pos. ✗ $\quad |1| > 0$

$\begin{vmatrix} 1 & 1 \\ 1 & 1 \end{vmatrix} = 0 \not> 0.$

$\Rightarrow$ No define prod. escalar.

(2) $u = (1,2,1)$
$v = (-1,1,1)$ $\in \mathbb{R}^3$.

a) Dado $u \cdot v = u_1v_1 + u_2v_2 + u_3v_3$ (usual de $\mathbb{R}^3$)

Hallar $\measuredangle(u,v)$:

$$\measuredangle(u,v) = \arccos\left(\frac{u \cdot v}{\|u\| \cdot \|v\|}\right) = \arccos\left(\frac{2}{\sqrt{6} \cdot \sqrt{3}}\right)$$

$$= \boxed{1.0799 \text{ rad.}}$$

$u \cdot v = 1 \cdot (-1) + 2 \cdot 1 + 1 \cdot 1 = 2.$

$\|u\| = \sqrt{u \cdot u} = \sqrt{u_1^2 + u_2^2 + u_3^2} = \sqrt{1^2 + 2^2 + 1^2} = \sqrt{6}.$

$\|v\| = \sqrt{v \cdot v} = \sqrt{v_1^2 + v_2^2 + v_3^2} = \sqrt{(-1)^2 + 1^2 + 1^2} = \sqrt{3}$

---

b) Con $u \cdot v = 3u_1v_1 + u_1v_2 + u_2v_1 + u_2v_2 + 2u_3v_3$

Hallar $\measuredangle(u,v)$:

$$\measuredangle(u,v) = \arccos\left(\frac{u \cdot v}{\|u\| \cdot \|v\|}\right) = \arccos\left(\frac{0}{\|u\| \cdot \|v\|}\right)$$

$$= \boxed{\frac{\pi}{2} \text{ rad}} \leftrightarrow (90^\circ)$$

$u \cdot v = 3 \cdot 1 \cdot (-1) + \ldots + 2 \cdot 1 \cdot 1 = 0$

---

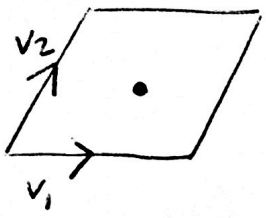
④ a) $L = \langle (1,-1,0), (5,3,-2) \rangle$ $\mathbb{R}^3$ usual.

v_1 v_2

- Antes de nada:  son l.i. ✓ — A ojo / Gauss



$\Rightarrow$ L es un



y $B = \{v_1, v_2\}$ base de L.

- ¿$L^\perp$?

Como $\dim L^\perp = \dim \underbrace{\mathbb{R}^3}_{\text{Ambiente}} - \dim L = 3 - 2 = \boxed{1}$

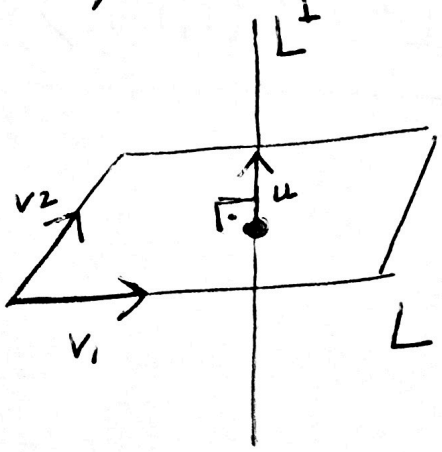
$\Rightarrow$ $L^\perp = \langle u \rangle$



y debe ser $\perp$ a L:

$\Leftrightarrow$

A sus generadores, l.i. que son los que "mandan".



Por tanto, si $u = (x,y,z) \in L^{\perp}$ debe ser

$$\begin{aligned} u \cdot v_1 &= 0 \\ u \cdot v_2 &= 0 \end{aligned} \iff \begin{aligned} (x,y,z)\cdot(1,-1,0) &= 0 \\ (x,y,z)\cdot(5,3,-2) &= 0 \end{aligned}$$

$$\iff \left.\begin{aligned} x - y &= 0 \\ 5x + 3y - 2z &= 0 \end{aligned}\right\} \equiv L^{\perp}.$$

Recordemos $L^{\perp}$ es <u>recta</u> (todas las letras a <u>una sóla</u>):

$$(x,y,z) \in L^{\perp} \Rightarrow (x,y,z) = \left(x, x, \tfrac{1}{2}(5x+3x)\right)$$

$$= (x, x, 4x)$$

$$= x(1,1,4)$$

$$\iff \underline{L^{\perp} = \langle (1,1,4) \rangle}.$$

- Ecs. Paramétricas ⇐⇒ Implícitas:

Si $L = \langle (1,-1,0), (5,3,-2) \rangle$, entonces:

$$(x,y,z) \in L \iff (x,y,z) = \lambda(1,-1,0) + \mu(5,3,-2)$$
$$= (\lambda+5\mu, -\lambda+3\mu, -2\mu)$$

$$\iff \begin{cases} x = \lambda+5\mu \\ y = -\lambda+3\mu \\ z = -2\mu \end{cases}$$ paramétricas de L.

¿Cuántas para implícitas?

$$\dim L = \dim \mathbb{R}^3 - (2g\,A)$$
$$2 \qquad 3 \qquad (1)$$

Sumando desde las paramétricas:

$$\begin{array}{l} x = \lambda+5\mu \\ + \quad y = -\lambda+3\mu \\ \quad z = -2\mu \end{array}$$

$$x+y+z = \cancel{x}+5\mu - \cancel{x}+3\mu-2\mu = 6\mu = 6\cdot\frac{z}{(-2)} = -3z$$

$$\iff \boxed{x+y+4z=0} \text{ implícita de } L.$$

- $B = \{v_1, v_2\} \xrightarrow{G\text{-}S} \widetilde{B} = \{\widetilde{u_1}, \widetilde{u_2}\}$

  base de L. base ortonormal de L.

Empezamos el proceso G-S:

1) $\widetilde{u_1} = \dfrac{v_1}{\|v_1\|} = \dfrac{(1,-1,0)}{\sqrt{1^2+(-1)^2+0^2}} = \boxed{\dfrac{1}{\sqrt{2}}(1,-1,0)}$.

2) $u_2 = v_2 - (v_2 \cdot \widetilde{u_1})\widetilde{u_1} = (5,3,-2) - \left((5,3,-2)\cdot\dfrac{1}{\sqrt{2}}(1,-1,0)\right)\dfrac{1}{\sqrt{2}}(1,-1,0)$

$= (5,3,-2) - \left(\dfrac{1}{\sqrt{2}}\right)(5-3+0)\cdot\left(\dfrac{1}{\sqrt{2}}\right)(1,-1,0)$

$= (5,3,-2) - \dfrac{1}{2}\cdot 2\,(1,-1,0)$

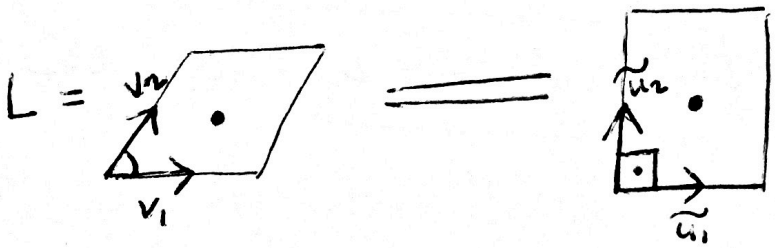
$= (5,3,-2) - (1,-1,0)$

$= (4,4,-2)$

$= 2(2,2,-1)$.

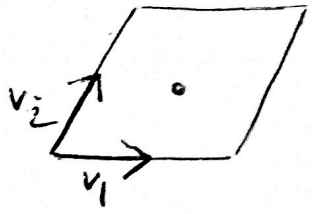
3) $\widetilde{u_2} = \dfrac{u_2}{\|u_2\|} = \boxed{\dfrac{1}{3}(2,2,-1)}$.

$\Rightarrow \widetilde{B} = \{\widetilde{u_1}, \widetilde{u_2}\}$ base ortonormal de L



c) $V = \langle (1,1,1), (1,0,1), (3,2,3) \rangle$
$\quad\quad\quad

— $V = \langle \underbrace{(1,1,1)}_{v_1}, \underbrace{(1,0,1)}_{v_2} \rangle$.

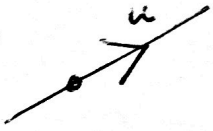


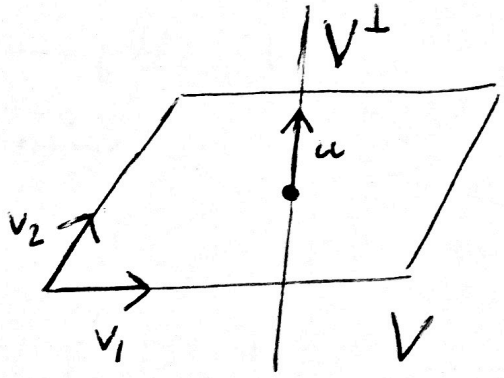
y $B = \{v_1, v_2\}$ base.

- ¿ $V^\perp$?

Tenemos que $\dim V^\perp = \dim \mathbb{R}^3 - \dim V$

$= 3 - 2 = \boxed{1}$,

luego $V^\perp = \langle u \rangle$  y además



ortogonal a V (a sus generadores).

Entonces,

si $u = (x,y,z) \in V^\perp \Rightarrow \begin{matrix} u \cdot v_1 = 0 \\ u \cdot v_2 = 0 \end{matrix}$

$\Leftrightarrow \left.\begin{matrix} x+y+z = 0 \\ x+z = 0 \end{matrix}\right\} \equiv V^\perp$

(Todas las letras a una sola).

Si $(x,y,z) \in V^{\perp} \Rightarrow (x,y,z) = (x,0,-x)$

$= x(1,0,-1)$

$\Leftrightarrow V^{\perp} = \langle (1,0,-1) \rangle$.

- Ecs. Par $\hookrightarrow$ Implícitas:

$V = \langle (1,1,1), (1,0,1) \rangle$.

Si $(x,y,z) \in V \Rightarrow (x,y,z) = \lambda(1,1,1) + \mu(1,0,1)$

$= (\lambda+\mu, \lambda, \lambda+\mu)$

$\Rightarrow \begin{cases} x = \lambda+\mu \\ y = \lambda \\ z = \lambda+\mu \end{cases}$ ecs. par. de V.

¿Cuántas implícitas?

$\dim V = \dim \mathbb{R}^3 - \text{rg } A$

2 3 ①

Sumo desde las paramétricas:

$$\begin{aligned} x &= \lambda+\mu \\ +\ y &= \lambda \\ z &= \lambda+\mu \end{aligned}$$

$$x+y+z = 3\lambda+2\mu$$

debe desaparecer. λ y μ.

Resto a la 1ª y 3ª:

$$\begin{aligned} x &= \lambda+\mu \\ -\ z &= \lambda+\mu \end{aligned}$$

$$x-z=0$$ ec. implícita

$y = \lambda \in \mathbb{R}$ es libre.

Es decir,

$$V = \{(x,y,z) \mid x-z=0\}.$$

- $B = \{v_1, v_2\}$ $\xrightarrow{G\text{-}S}$ $\widetilde{B} = \{\widetilde{u}_1, \widetilde{u}_2\}$

  base de V — base orto-normal de V.

Empezamos el proceso GS:

1) $\widetilde{u}_1 = \dfrac{v_1}{\|v_1\|} = \dfrac{(1,1,1)}{\sqrt{1^2+1^2+1^2}} = \boxed{\dfrac{1}{\sqrt{3}}(1,1,1)}$.

2) $u_2 = v_2 - (v_2 \cdot \widetilde{u}_1)\widetilde{u}_1 = (1,0,1) - \left((1,0,1)\cdot\frac{1}{\sqrt{3}}(1,1,1)\right)\frac{1}{\sqrt{3}}(1,1,1)$

$= (1,0,1) - \frac{1}{\sqrt{3}}\cdot 2 \cdot \frac{1}{\sqrt{3}}(1,1,1)$

$= (1,0,1) - \frac{2}{3}(1,1,1)$

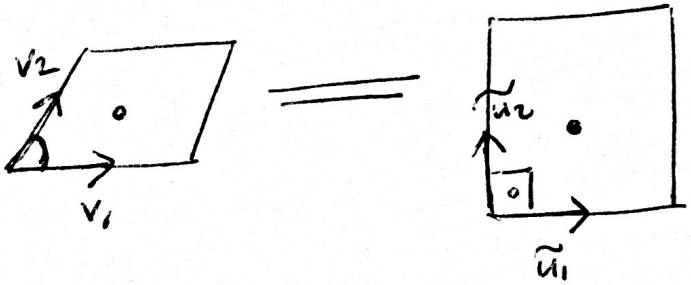
$= \left(\frac{1}{3}, \frac{-2}{3}, \frac{1}{3}\right)$

$= \frac{1}{3}(1,-2,1)$.

3) $\widetilde{u}_2 = \dfrac{u_2}{\|u_2\|} = \dfrac{\frac{1}{3}(1,-2,1)}{\sqrt{\left(\frac{1}{3}\right)^2+\left(\frac{-2}{3}\right)^2+\left(\frac{1}{3}\right)^2}} = \boxed{\dfrac{1}{\sqrt{6}}(1,-2,1)}$

$\Rightarrow \widetilde{B} = \{\widetilde{u}_1, \widetilde{u}_2\}$ base orto-normal de V.

$V =$



(5) $V = \{(x,y,z) \mid x+y+z=0\}$.

¿ $\widetilde{B}$ orto-normal de V ?

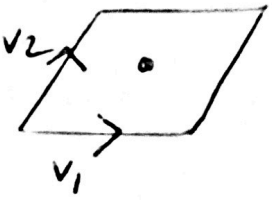
Como antes, necesitamos que V venga como $V = \langle \dots \rangle$.

(li)

Es decir, necesitamos BASE de V !

Empezamos:

- ¿Qué es V?



$\Rightarrow$ $V = \langle v_1, v_2 \rangle$ y $B = \{v_1, v_2\}$ será base.
li

3 letras $\rightsquigarrow$ 2.

Si $(x,y,z) \in V \Rightarrow (x,y,z) = (x,-x-z,z)$

$= (x,-x,0) + (0,-z,z)$

$= x(1,-1,0) + z(0,-1,1)$

$\Leftrightarrow V = \langle \underbrace{(1,-1,0), (0,-1,1)} \rangle$

li ✓

$\Rightarrow$ $B = \{(1,-1,0), (0,-1,1)\}$ base de V
$\quad\quad\quad

⑥ Pares ortogonales.

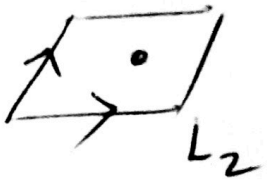
(¿Lo son sus generadores?)

li

a) $L_1 = \langle (-3,0,1) \rangle$

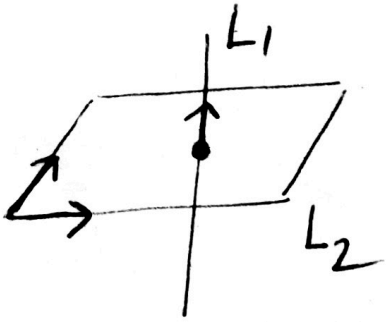


$L_2 = \{(x,y,z) \mid 3x - z = 0\}$



Me preguntan si $L_1 \perp L_2$:

¿



?

Me basta ver si lo son sus generadores li.

Para eso hace falta poner $L_2 = \langle v_1, v_2 \rangle$.

$L_2 =$  $\Leftrightarrow$ $(x,y,z) \in L_2$

$= (x,y,3x) = (x,0,3x) + (0,y,0)$

$= x(1,0,3) + y(0,1,0)$

$\Leftarrow$ $L_2 = \langle \underbrace{(1,0,3),(0,1,0)} \rangle$

li.

Por todo,

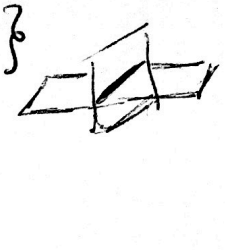
$L_1 \perp L_2 \Leftrightarrow$ $(-3,0,1)\cdot(1,0,3) = 0$ ✓

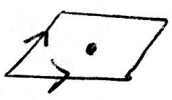
$(-3,0,1)\cdot(0,1,0) = 0$ ✓.

$\mathbb{R}^3$

b) $V_1 = \{(x,y,z) \mid x+y-z=0,\ x-y=0\}$

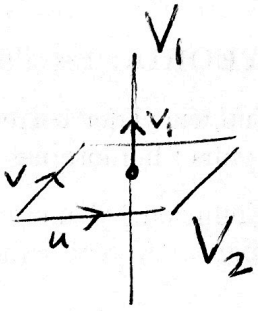
$V_2 = \{(x,y,z) \mid x+y=0\}$.





$V_1 = \langle v_1 \rangle$, $V_2 = \langle u, v \rangle$ l.i.

¿ $V_1 \perp V_2 \iff v_1 \perp u, v$?



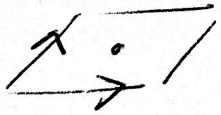
- $V_1 \equiv \begin{cases} x+y-z=0 \\ x-y=0. \end{cases}$



* $(x,y,z) \in V_1 \Rightarrow (x,y,z) = (x, x, 2x)$

$= x(1,1,2)$

$\iff \boxed{V_1 = \langle (1,1,2) \rangle}$

- V_2  $(x,y,z) \in V_2 \Rightarrow (x,y,z) = (x, -x, z)$

La z no está: se pone

$\Rightarrow V_2 = \langle (1,-1,0), (0,0,1) \rangle$ l.i.

$= (x,-x,0) + (0,0,z)$

$= x(1,-1,0) + z(0,0,1)$

Entonces:

$(1,1,2) \cdot (1,-1,0) = 0$ ✓

$(1,1,2) \cdot (0,0,1) = 2 \neq 0$

$\Rightarrow \boxed{V_1 \not\perp V_2}$.

X

⑧ $D \equiv \begin{cases} x+2y-z-2t=0 \\ 2x+y-2z-t=0 \\ 2x+7y-2z-7t=0. \end{cases}$ $\mathbb{R}^4$

- ¿Qué es D?

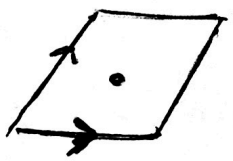
$$\dim D = \dim \mathbb{R}^4 - \operatorname{rg}\begin{pmatrix} 1 & 2 & -1 & -2 \\ 2 & 1 & -2 & -1 \\ 2 & 7 & -2 & -7 \end{pmatrix}$$

$$= 4 - \operatorname{rg}\begin{pmatrix} 1 & 2 & -1 & -2 \\ 0 & -3 & 0 & 3 \\ 0 & 0 & 0 & 0 \end{pmatrix} = 4-2 = \boxed{2}$$

$$\begin{pmatrix} 1 & 2 & -1 & -2 \\ 2 & 1 & -2 & -1 \\ 2 & 7 & -2 & -7 \end{pmatrix} \overset{F_2 \leftrightarrow F_2 - 2F_1}{\sim} \begin{pmatrix} 1 & 2 & -1 & -2 \\ 0 & -3 & 0 & 3 \\ 2 & 7 & -2 & -7 \end{pmatrix}$$

$$\overset{F_3 \leftrightarrow F_3 - 2F_1}{\sim} \begin{pmatrix} 1 & 2 & -1 & -2 \\ 0 & -3 & 0 & 3 \\ 0 & 3 & 0 & -3 \end{pmatrix} \leftarrow \leftarrow$$

$$\overset{F_3 \leftrightarrow F_3 + F_2}{\sim} \begin{pmatrix} 1 & 2 & -1 & -2 \\ 0 & -3 & 0 & 3 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

$\Rightarrow$ D es un  en $\mathbb{R}^4$

$\Rightarrow$ $D = \langle u, v \rangle$ l.i., $B = \{u, v\}$ base de D.

Si $(x,y,z,t) \in D \Rightarrow (x,y,z,t) = (x,y,x,y)$

$= (x,0,x,0) + (0,y,0,y)$

$= x(1,0,1,0) + y(0,1,0,1)$.

De la última matriz,

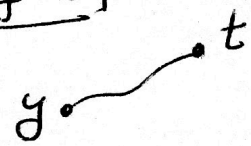
$$\begin{pmatrix} 1 & 2 & -1 & -2 \\ 0 & -3 & 0 & 3 \\ 0 & 0 & 0 & 0 \end{pmatrix} \rightsquigarrow \begin{array}{l} x+2y-z-2t=0 \\ -3y+3t=0 \end{array}$$

$\Rightarrow \boxed{y=t}$

$\Rightarrow x+2\cancel{y}-z-2\cancel{y}=0$

$\boxed{x=z}$





$\Leftarrow$ $D = \langle \overbrace{(1,0,1,0)}^{v_1}, \overbrace{(0,1,0,1)}^{v_2} \rangle$.

l.i.

ie. $B = \{v_1, v_2\}$ base D.

a) Ecs. Par. de D.

Las obtenemos a partir de los generadores de una base:

$$(x,y,z,t) \in D \Rightarrow (x,y,z,t) = \lambda(1,0,1,0) + \mu(0,1,0,1)$$
$$= (\lambda,0,\lambda,0) + (0,\mu,0,\mu)$$
$$= (\lambda,\mu,\lambda,\mu)$$

$$\Rightarrow \begin{cases} x=\lambda \\ y=\mu \\ z=\lambda \\ t=\mu \end{cases} \iff \begin{cases} x=z=\lambda. \\ y=t=\mu. \end{cases}$$

$$B = \{\underbrace{(1,0,1,0)}_{v_1}, \underbrace{(0,1,0,1)}_{v_2}\} \xrightarrow{G\text{-}S} \widetilde{B} = \{\widetilde{u}_1, \widetilde{u}_2\}.$$

Seguimos los pasos de G-S: $\mathbb{R}^4$ usual.

1) $\widetilde{u}_1 = \dfrac{v_1}{\|v_1\|} = \dfrac{(1,0,1,0)}{\sqrt{2}} = \boxed{\frac{1}{\sqrt{2}}(1,0,1,0)}$

2) $u_2 = v_2 - (v_2 \cdot \widetilde{u}_1)\widetilde{u}_1 = (0,1,0,1) - (0)\cdot\widetilde{u}_1 = (0,1,0,1) = v_2$

3) $\widetilde{u}_2 = \dfrac{u_2}{\|u_2\|} = \dfrac{(0,1,0,1)}{\sqrt{2}} = \boxed{\frac{1}{\sqrt{2}}(0,1,0,1).}$

$\Rightarrow \widetilde{B} = \{\widetilde{u}_1, \widetilde{u}_2\}$ base ortonormal.

b) $\text{proy}_D u = \text{proy}_{\tilde{B}} u = \text{proy}_{\langle \tilde{u}_1, \tilde{u}_2 \rangle} u$!

proy ⊥ sobre bases ortonormales $\tilde{B}$

$$= (u \cdot \tilde{u}_1)\tilde{u}_1 + (u \cdot \tilde{u}_2)\tilde{u}_2$$

$$= \frac{1}{\sqrt{2}}(2+3)\tilde{u}_1 + \frac{1}{\sqrt{2}}(-2-3)\tilde{u}_2$$

$$= \frac{5}{\sqrt{2}}\tilde{u}_1 - \frac{5}{\sqrt{2}}\tilde{u}_2$$

$$= \frac{5}{\sqrt{2}}(\tilde{u}_1 - \tilde{u}_2) = \boxed{\frac{5}{2}(1,-1,1,-1)}$$

$u = (2,-2,3,-3)$

$\tilde{u}_1 = \frac{1}{\sqrt{2}}(1,0,1,0)$

$\tilde{u}_2 = \frac{1}{\sqrt{2}}(0,1,0,1)$

c)

$\mathbb{R}^4$

$B = \{e_1, e_2, e_3, e_4\}$ ortonormal

Reunir las 2 en 1

D

$\tilde{B} = \{\tilde{u}_1, \tilde{u}_2\}$

ortonor-al

$D^\perp$

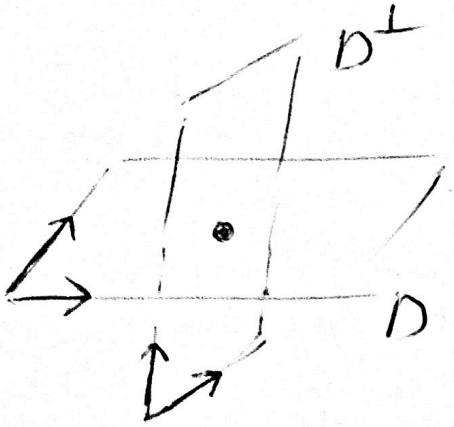
$\tilde{B}' = \{\tilde{u}_3, \tilde{u}_4\}$

ortonor-al

Halla-os $D^\perp$

$\dim D^\perp = \dim \mathbb{R}^4 - \dim D = 4 - 2 = 2$

$\Rightarrow D^\perp$ $v \perp u$, $D^\perp = \langle u, v \rangle$ li



¿ Debemos empezar de D para encontrar u, v?

Sabemos que debe

- generar l.i. todo $D^\perp$
- ser $\perp$ a $\tilde{u}_1$ y $\tilde{u}_2$.

Entonces:

$$\tilde{u}_1 = \frac{1}{\sqrt{2}}(1,0,1,0) \rightsquigarrow u = (1,0,-1,0)$$

$$\tilde{u}_2 = \frac{1}{\sqrt{2}}(0,1,0,1) \rightsquigarrow v = (0,1,0,-1).$$

¿ Qué cumplen u y v?

$u \perp \tilde{u}_1, \tilde{u}_2$ ✓

$v \perp \tilde{u}_1, \tilde{u}_2$ ✓

$u \perp v$ ✓

$\Rightarrow$ Nos sirven como generadores l.i

$D^\perp = \langle u, v \rangle$.

Lo que pasa es que no tienen $\|\ \| = 1$ para formar base ortonormal (sí son $\perp$ entre sí).

Entonces los normaliza-os: (*)

$$u \rightsquigarrow \tilde{u}_3 = \frac{u}{\|u\|} = \boxed{\frac{1}{\sqrt{2}}(1,0,-1,0)}$$

$$v \rightsquigarrow \tilde{u}_4 = \frac{v}{\|v\|} = \boxed{\frac{1}{\sqrt{2}}(0,1,0,-1)}$$

Ahora sí:

$\tilde{B} = \{\tilde{u}_1, \tilde{u}_2, \tilde{u}_3, \tilde{u}_4\}$ base ortonormal de $\mathbb{R}^4$

$\perp$ 2 a 2 ✓

$\|\ \| = 1$ ✓

(*) Nota. Recuerdo que si v es un vector,

$$u = \frac{v}{\|v\|}$$ siempre es unitario ($\|\ \| = 1$).

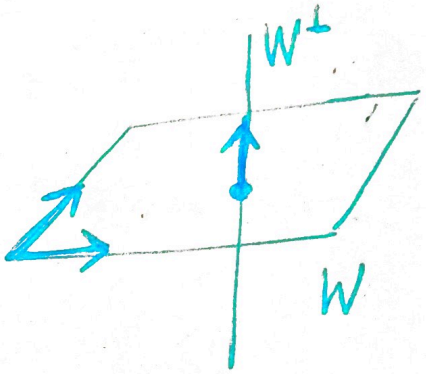
(11) a) $W = \langle (1,0,1), (2,-1,1) \rangle$ $\quad \mathbb{R}^3$ usual.

l.i ✓

¿$W^\perp$?

$\dim W^\perp = \dim \mathbb{R}^3 - \dim W = 3 - 2 = \boxed{1}$

$\Rightarrow$ $W^\perp = \langle w \rangle$ $\quad v = (x,y,z)$.



Si $(x,y,z) \in W^\perp$,

$$\left.\begin{array}{l} (x,y,z)\cdot(1,0,1) = 0 \\ (x,y,z)\cdot(2,-1,1) = 0 \end{array}\right. \Leftrightarrow \left.\begin{array}{l} x+z=0 \\ 2x-y+z=0 \end{array}\right\} \equiv W^\perp$$

Luego (todo a una letra):

$$(x,y,z) \in W^\perp \Rightarrow (x,y,z) = (x, x, -x)$$
$$= x(1,1,-1)$$
$$\Leftrightarrow W^\perp = \langle (1,1,-1) \rangle.$$

b) $W = \{(x,y,z,t) \mid x-y+z+t=0,\ 2x-y=0\}$

$$\equiv \begin{cases} x-y+z+t=0 \\ 2x-y=0 \end{cases} \qquad \mathbb{R}^4. \qquad ¿W^\perp?$$

$$\dim W^\perp = 4 - \dim W = 4-2 = (2).$$

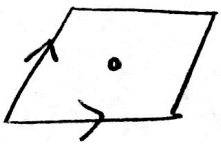
$$\dim W = \dim \mathbb{R}^4 - \text{rg}\begin{pmatrix} 1 & -1 & 1 & 1 \\ 2 & -1 & 0 & 0 \end{pmatrix}$$

$$= 4-2 = \boxed{2}$$

$$\begin{pmatrix} 1 & -1 & 1 & 1 \\ 2 & -1 & 0 & 0 \end{pmatrix} \underset{F_2 \leftrightarrow F_2 - 2F_1}{\sim} \begin{pmatrix} 1 & -1 & 1 & 1 \\ 0 & 1 & -2 & -2 \end{pmatrix} \checkmark$$

$$\underset{F_1 \leftrightarrow F_2 + F_1}{\sim} \begin{pmatrix} 1 & 0 & -1 & -1 \\ 0 & 1 & -2 & -2 \end{pmatrix}.$$

$\Rightarrow$ $W^\perp = \langle u, v \rangle$ l.i.



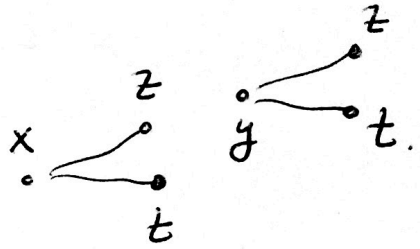
Necesito

$$W = \langle w_1, w_2 \rangle$$

l.i.

De la última matriz:

$$\begin{pmatrix} 1 & -1 & 1 & 1 \\ 0 & 1 & -2 & -2 \end{pmatrix} \rightsquigarrow \begin{array}{l} x-y+z+t=0 \\ y-2z-2t=0 \end{array}$$



$$\begin{array}{l} x-y+z+t=0 \\ y-2z-2t=0 \end{array}$$

$x-z-t=0 \Rightarrow \boxed{x=z+t}$.

$y-2z-2t=0 \Rightarrow \boxed{y=2z+2t}$.

Luego

$$(x,y,z,t)\in W \Rightarrow (x,y,z,t)=(z+t,\ 2z+2t,\ z,\ t)$$

$$= (z,\ 2z,\ z,\ 0)+(t,\ 2t,\ 0,\ t)$$

$$= z(1,2,1,0)+t(1,2,0,1).$$

$$\Leftarrow W=\langle (1,2,1,0),(1,2,0,1)\rangle .$$

Ahora calculamos $W^{\perp}$:

$$(x,y,z,t)\in W^{\perp} \Rightarrow \begin{array}{l} (x,y,z,t)\cdot(1,2,1,0)=0 \\ (x,y,z,t)\cdot(1,2,0,1)=0 \end{array}$$

$$\Leftarrow W^{\perp} \equiv \begin{cases} x+2y+z=0 \\ x+2y+t=0 \end{cases}$$

$$x+2y+z=0 \quad \Rightarrow \quad z=-x-2y$$

$$-\ x+2y+t=0$$

$$z-t=0 \Rightarrow \boxed{z=t}$$

$$(x,y,z,t)=(x,y,-x-2y,-x-2y)$$

$$\cdots$$

$$W^{\perp}=\langle (1,0,-1,-1),(0,1,-2,-2)\rangle .$$